Mini
Math
Zoo
Count & Care
ZOO
5=5?
1
2
3

GAME DESIGN BOOK

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Genre : Children's Educational / Early Math Learning / Puzzle

Platform : Mobile / Pc

Sort & Count Game Design AssignmentSort

This document outlines my design approach for elevating the "Sort & Count" prototype into an engaging educational game for 5yearolds (and potentially older kids/teens/adults with scalable difficulty). Based on my concept of a zoo management adventure, I've integrated exploration and fun while strictly adhering to the locked mechanics: Sort -> Count -> Compare as a 2D draganddrop experience. No mechanics have been redesigned; instead, the story and visuals wrap around themto make math feel organic and exciting. I've drawn from my observations of kids preferring exploratory, fun games over rote tasks, so the zoo world encourages curiosity without adding new gameplay loops. Annotations explain my thinking, emphasizing pedagogy for prereaders, motor skills, and empathy for young players.

Step 1: Concept & Story

World Overview
Sort & Count takes place in 'Sunshine Zoo', a vibrant training zoo designed especially for Junior Zookeepers. This is a living, breathing world where animals move, play, and depend on the player to maintain order. Counting is not an abstract task it is how the zoo functions. If animals are not sorted correctly, habitats become messy. If animals are not counted properly, someone could be missing.

Core Fantasy
The player is not a student completing worksheets. They are the youngest official Zookeeper in training. They wear a badge. They carry a Zoo Logbook. The zoo only opens when they finish their daily responsibilities.

Story Hook
Each morning begins with light chaos. Animals have wandered. Tigers might be near elephants. Monkeys might be in the wrong trees. Pip arrives in a flutter, explaining that the zoo gates cannot open until everything is organized and counted.

Who is Pip?
Pip is a baby parrot and the assistant to the Head Zookeeper. He is energetic, curious, and slightly dramatic. He cannot count confidently yet which makes the child the real expert. Pip reacts emotionally: excited when things are correct, worried when numbers don't match. He guides through gestures and expression rather than heavy text.

Why Sorting -> Counting -> Comparing Matters
Sorting ensures animals are safe in the correct habitats. Counting ensures no animal is missing. Comparing ensures fairness equal food portions, equal space, and safety balance. Math becomes responsibility. Responsibility becomes pride.

Stakes & Emotional Arc
If numbers don’t match, Pip grows concerned. The player explores to find missing animals. When the final comparison is correct, the zoo gates open, visitors cheer, and animals celebrate. The player earns a golden badge stamp in their Logbook. The child leaves the session feeling capable, observant, and important not because they solved math, but because they protected and helped animals.

2. Visual Design / Style Treatment

Dominant Visual Tone

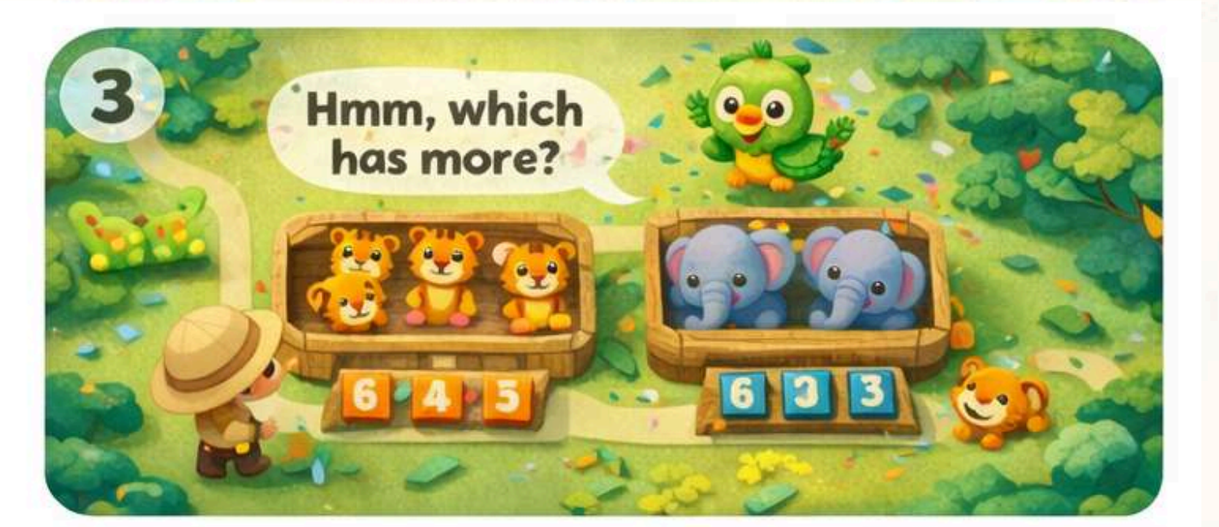
Soft 2D storybook illustration style with rounded silhouettes and gentle shadows. Everything feels safe, warm, and touchable. No sharp corners. No harsh contrast. Shapes are chunky and readable.

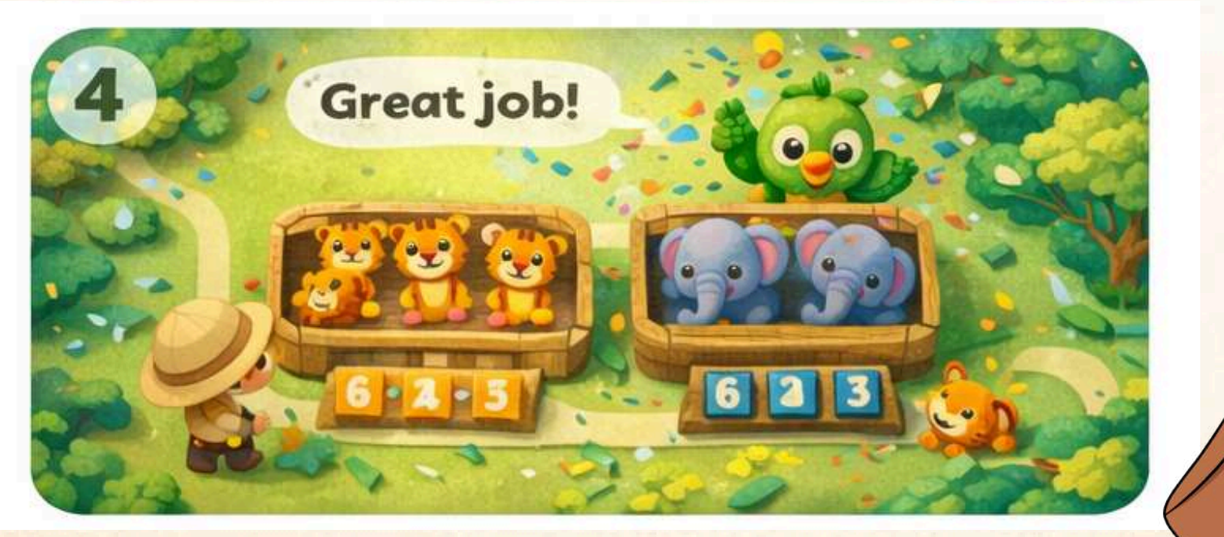
Mood Board Direction

1. Storybook zoo illustrations
2. Watercolor texture backgrounds
3. Clay-like rounded characters
4. Big expressive cartoon eyes
5. Paper-cut layered landscapes
6. Wooden toy-inspired UI elements
7. Warm sunrise lighting
8. Sticker style interactive buttons

1
Sort the animals!
37
Léto
4 3
2 5

2
27
Léto
4 2 5
2 3 3


3
Hmm, which has more?
6 4 5
6 3 3


4
Great job!
6 4 5
6 3 3

Color Palette & Emotional Intent

Jungle Green – #4CAF50 (Safety, nature, growth)
Sunny Yellow – #FFD54F (Energy, happiness, reward)
Sky Blue – #64B5F6 (Calm guidance)
Soft Coral – #FF8A65 (Attention & excitement)
Cream White – #FFF8E1 (Soft background balance)

Typography Scale

Primary Font Style: Rounded, pre-reader friendly sans serif.

40pt – Headline Example: 'Mini Math Zoo'.
28pt – Prompt Example: 'Let's Count the Tigers!'
22pt – UI Labels (Icon-supported)(cartooned typed kids attracted

Text is always paired with visual icons to support pre-readers.

Pip Character Concept

Size: Small enough to feel like a companion, large enough for readable expressions.
Color: Bright green body, yellow chest, coral beak.

Expression 1 – Celebration: Wings spread, eyes wide, sparkles around him.
Expression 2 – Thinking: Head tilted, one eyebrow raised.
Expression 3 – Concerned: Slight frown when numbers don't match.

Environment Design

The background shifts between themed zones: Jungle, Savanna, and Water Habitat. Enclosures are circular habitats integrated into the world not UI boxes. Fences are rounded wood posts. Paths are curved and soft.

3. UI Design / Pedagogy First Layout

SORT Screen Annotated Design Thinking

- Animals roam in center play area.
- Habitats are large circular zones with visual texture (grass, sand, water).
- Drop zones glow when animal is nearby (magnetic assistance).
- Incorrect placements gently bounce back (no punishment).
- Pip floats top-right but flies closer during mistakes.
- Goal is visually implied: empty habitats subtly pulse.

COUNT Screen – Physical Counting Design

- Animals auto-align into neat rows for clarity.
- Player taps each animal to generate a wooden tally token.
- Each tap produces a satisfying click sound.
- Count builds physically on screen (stacked tokens).
- Large numeral appears only after tapping sequence.

COMPARE Screen – Spatial Understanding

- Two habitats displayed side-by-side.
- Animals auto-space evenly in rows.
- Larger group slightly stretches wider.
- Equal groups glow symmetrically.
- Player selects from 3 giant buttons: More / Fewer / Equal (icon-based).
- Pip gestures but never reveals answer.

Motor Skill & Accessibility Choices

- Oversized interaction zones
- Snap-to-grid drop magnetism
- Forgiving drag release detection
- Minimal small UI elements
- Clear color-coded grouping

Feedback & Reward System

Correct actions trigger environmental celebration: animals cheer, balloons rise, sunlight brightens, and the zoo gate slowly opens.
Completion grants a golden Logbook stamp.
Progression feels like unlocking responsibility, not unlocking levels.

Design Philosophy & Studio Alignment

This concept elevates a simple drag-and-drop mechanic into a responsibility-driven adventure. It respects 5-year-olds as capable thinkers. It designs for prereaders. It makes math meaningful through narrative purpose.

The child is not completing tasks they are running a zoo.

THANK YOU